

RUCHIKA SHARMA

Email - ruchika16195@gmail.com

Contact - 91-7062626323

LinkedIn - [Check out my LinkedIn profile](#)

Professional Summary

Data Engineer, with strong interpersonal skills having more than 5 years of experience in dealing with Technical aspects of Healthcare Data.

Areas of Interest: Developing analytical modules, extracting scientific insights, adding value to Precision Medicine, Developing & Maintaining huge scientific data resources and hubs, Developing Product Workflows and Project Management. Dealing with Real world problems, Evidences, streaming data and its corresponding outcomes.

Skills

Data Engineering & Data Analytics: ● AWS Services CFT, CI/CD, ETL processing ● Airflow, DMS, AWS Glue, Redshift. ● Cloud computing, ECS, EKS, EMR clusters, Apache Kafka, and processing streaming data. ● Statistical analysis: (PCA, mean, median, z-scores, Log2FC, Hypothesis testing) ● Data Normalization: Combat, SVA, UMAP, and TSNE ● Fixing data issues, removing redundant information, aggregating data by considering its scientific aspects	Technical Skills: ● Programming In Python, SQL ● Cloud Computing, handling EC2 instances, Docker, Github and version controlling ● Working with Step Function, Lambda, AWS Batch, S3, Jenkins. ● Handling distributed file systems and multiprocessing ● Handling and Maintaining Databases, MongoDB, Postgres, Oracle. ● JIRA - Backlog screening and grooming, User story generation and task allocation
Bioinformatics: ● Computational biology (Microarray, NGS, and MultiOmics) ● Variant Analysis and Prioritization ● Parsing & Processing different file formats from NGS data ● Pathway-Based Network and Gene Co-Expression Analysis ● Configuring and Executing WDL pipelines and Nextflow workflows according to client requirements.	Soft Skills: ● Networking and Relationship building. ● Critical thinking and problem solving ● Inspiring and motivating others and taking responsibility for outcomes ● Contributing positively to company culture and driving workplace satisfaction with work/life balance.

Experience (5+ Years)

Data Engineer - Bioinformatics at Eli Lilly and Company (Sept 2024 Onwards)

- Adjusting model weights and enhancing bot performance. Combining multiple bots to create a cooperative multi agent system to support clients with queries related to health, safety and environment.
- Developing and Deploying Airflow DAGs, Preparing ETL workflows to capture data from different in-vivo, in-vitro and omics data sources for assay based data products.

Bioinformatics Scientist at ConcertAI (Sept 2023 - July 2024) - Preparing and implementing workflows:

- Reducing manual validation of curated concepts (like biomarker, RNA Transcript and variants) up to 90%.
- Increasing Product Integrity and Quality, by Normalising and Transforming Data into Standard formats and naming structure.
- Configuring the data curation modules according to biological concepts.

PierianDx - Velsera (April 2022 - Aug 2023) - Developing Bioinformatic tools and standalones:

- Implementing software workflows, Impact analysis to changes in the modules,
- Preparing and optimizing AWS batch jobs and workflows
- Handling differences in variant annotations due to different Refseq versions.
- Maintaining Agile documentations and methodologies w.r.t software maintenance and upgradation

Innoplexus Consulting Services (June 2019 to April 2022) - Developing Genomics Pipelines

- Analyzing diverse multi-layered omics data to identify novel drug targets and biomarkers. Mutation Burden Profiling for Precision Medicine and Patient Stratification.
- Developing a complete NGS multi-omics pipeline from start to end with customized features. Growing it into a user-friendly Dashboard.

Internships & Projects

NIMHANS, Bangalore ADBS (Accelerated program for Discovery in Brain disorders using Stem cells) 6 months. Whole Exome sequencing Data Analysis, Risk variant determination and Mutation Burden Analysis. Mutation Burden Profile of Lipid Metabolism in 18 Familial Alzheimer's disease Cases, in AAIC 2018.

Bennett University (March 2019 to June 2019) - Scientific research work and findings on extinct medicinal plants to determine enzymatic inhibitors of valued secondary metabolites involved using NGS Transcriptomes.

Education

University of Rajasthan – **Integrated M.TECH (Biotechnology & Bioinformatics)**

August 2017 - July 2018 | 72.6% (Aggregate)

University of Rajasthan - **B.TECH (Biotechnology & Bioinformatics)**

July 2013 - June 2017. | 67.2% (Aggregate)